LiDAR Point Cloud Denoising for Autonomous Driving Applications

Final Technical Report

Capstone Project CPG No. 5
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Abstract

Adverse weather degrades automotive LiDAR through two distinct physical mechanisms: airborne droplets scatter the beam and create spurious returns, while attenuation destroys weak returns outright. This project implements and compares six denoising methods on KITTI scans corrupted by a physics-based rain model, and evaluates them not only by reconstruction error but by their effect on a downstream 3D object detector.

Three results are reported. First, a lightweight DGCNN closes 98.4% of the *recoverable* reconstruction error — measured against an oracle that performs perfect noise classification — but requires 1.8 s per frame. Second, and most consequential, **Chamfer Distance does not predict downstream detection quality**: a PointNet autoencoder improves Chamfer Distance by 8.9% over the corrupted input while destroying 76% of the detector's recall ($p < 10^{-4}$), and the metric is separately shown to be improvable simply by emitting more points. Third, moving to a range-image representation yields *RangeRestoreNet*, which attains the best noise-removal F1 of any method evaluated (0.814) at 45 ms per frame — a 40× speedup that meets the project's 10 Hz real-time target with a learned model for the first time.

Two negative results are reported with equal prominence. Position-restoration heads — present in both learned models — contribute nothing, for a reason that is intrinsic rather than a training failure. And a self-supervised variant trained on physics-derived pseudo-labels reproduces its supervision rather than improving on it, quantifying exactly what ground-truth labels are worth on this task. A section on experimental validity documents seven defects found during review, including one that reversed a headline claim once tested at adequate statistical power.

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1 Introduction

1.1 Problem

LiDAR is an active sensor: it emits laser pulses and measures the time of flight of their returns. That is precisely why weather degrades it. Anything suspended in the air between the sensor and its target — raindrops, fog droplets, snowflakes — interacts with the beam, and it does so in two ways that must be modelled separately:

- **Backscatter.** A droplet intercepting the beam reflects it before it reaches the intended target. The sensor reports a return at the droplet's range, producing a *phantom point* in mid-air that no solid surface generated.
- **Attenuation.** Energy is absorbed and scattered along the path, so returns from distant or weakly reflective surfaces fall below the detection threshold and are *lost entirely*.

The second mechanism is the one most often overlooked, and it governs how every result in this report must be read. A denoiser can delete a spurious point and it can reposition a displaced one, but it cannot reconstruct a return the weather destroyed. Reconstruction error therefore has a strictly positive floor, quantified in Section 3.5.

1.2 Objectives

Taken from the project proposal, Section 5:

1. Preprocess raw LiDAR scans and model adverse-weather corruption.
2. Implement and compare statistical and learning-based denoisers.
3. Evaluate reconstruction quality *and* downstream perception impact.
4. Optimise toward the 10 Hz scan rate of automotive LiDAR.

1.3 Contributions

1. A six-method comparison on a common, statistically controlled protocol, including two methods drawn from 2025–2026 literature and one developed here.
2. An *oracle-gap* evaluation protocol that expresses each method's error as a fraction of the error that is actually recoverable, rather than against an unreachable zero.
3. Quantitative evidence, at $p < 10^{-4}$, that reconstruction metrics fail to predict downstream detection performance — with a second, independent demonstration that Chamfer Distance can be improved by densification alone.
4. *RangeRestoreNet*, a dual-head range-image denoiser reaching 22 Hz — the only learned method here that meets the real-time target.
5. Two documented negative results, with mechanisms identified.

2 Related Work

The adverse-weather LiDAR denoising literature divides cleanly by representation and by task formulation.

Representation. Early learned methods operate on raw points (PointCleanNet, score-based denoising), but every recent method targeting real-time operation projects the scan into a 2-D range image and applies convolutions: WeatherNet, LiSnowNet, 4DenoiseNet [3], LRINet [7], LIORNet [5]. The reason is throughput. A rotating LiDAR is natively a 2-D array of (laser channel, azimuth) samples, so the projection recovers the sensor's own topology, turning neighbourhood queries into array indexing. Reported speeds range from 16 Hz (TripleMixer [4]) to 43 Hz (LIORNet); the point-based method implemented here manages 0.55 Hz. This gap motivated Section 3.4.

Task formulation. Essentially all of this work treats denoising as *point-wise binary classification* — emit a noise mask, score it with IoU. The August 2025 survey [8] taxonomises denoising into "outlier removal" (dropping points) and "surface noise removal" (displacing points toward the underlying surface), and files every adverse-weather method under the former, noting that outlier removal "can only drop noisy points without adjusting their positions". Whether that limitation binds in practice is examined in Section 6; the answer found here is that it does not, for a reason the survey does not discuss.

Corruption models. Since no dataset records the same scene with and without rain, corruption is simulated. LISA [1] provides a physically-derived model based on Mie scattering, and Robo3D [2] established a three-level severity convention. Both are adopted here.

What this report adds. The survey does not critique Chamfer Distance as a downstream proxy — it treats CD and point-to-mesh distance as complementary. The evidence in Section 4.3 suggests that is too generous, and that a downstream task metric is not optional.

3 Methodology

3.1 Data

262 scans from two public KITTI raw drives (2011_09_26, drives 0001 and 0005), split 213/26/28 into train/validation/test *by frame* with a fixed seed. The official KITTI 3D-object Velodyne archive (29 GB, login-gated) was deliberately not used: the raw split provides identical sensor data at 490 MB, which is sufficient for denoising experiments.

Raw scans carry no ground-truth 3D boxes. This is the reason detection is evaluated by the *consistency protocol* of Section 3.5 rather than by absolute mAP, and the limitation is acknowledged rather than worked around.

Preprocessing chains NaN/Inf removal, a 60 m range gate, and 0.15 m voxel-grid downsampling. **The intensity channel is retained throughout.** This matters: weather backscatter returns markedly less energy than a solid surface at the same range — measured on this data, real returns have median intensity 0.150 against 0.051 for injected noise — and two of the six methods depend on it.

3.2 Weather model

Following LISA [1], the extinction coefficient for rain rate R (mm/h) at 905 nm is

$$\alpha = 0.003\, R^{0.6}\ \mathrm{m}^{-1}, \qquad T(r) = e^{-2\alpha r},$$

where $T(r)$ is the two-way transmittance at range r. Each return is then subjected to three processes:

1. **Backscatter.** With probability $0.35\,(1 - T(r))$ a droplet intercepts the beam; the return relocates to the droplet's range, drawn from $\mathrm{Exp}(1/2\alpha)$ truncated to $(0.5, r)$.
2. **Loss.** Non-scattered returns survive with probability $T(r)^{1/2}$; the remainder fall below the detection threshold and are deleted.
3. **Jitter.** Survivors receive SNR-dependent range noise, plus Gaussian sensor jitter ($\sigma = 0.05$ m) and 8% uniform outliers modelling multi-echo interference.

Intensity follows the same radiometry: a surviving surface return carries $i \cdot T(r)$, while a droplet return is weak for two independent reasons — a raindrop's backscatter cross-section is far smaller than a surface's, and its path is shorter. Three severity presets (10/25/50 mm/h) follow the Robo3D convention; all headline results use *moderate*, 25 mm/h.

At that severity the model destroys **31% of returns outright**. This single number governs the interpretation of every result below.

3.3 Methods

Six denoisers are compared. Three require no training:

SOR+ROR The proposal's statistical baseline. Statistical Outlier Removal ($k = 20$, $\alpha = 2.0$) removes points whose mean k-NN distance exceeds $\mu + \alpha\sigma$; Radius Outlier Removal then drops points with fewer than 4 neighbours within 0.6 m.

IDSOR A reimplementation of [6]. Plain SOR applies one global threshold, which is wrong in two opposite directions at once. Angular resolution is fixed, so genuine returns grow further apart with range and a global threshold erodes far-field structure; separately, weather backscatter returns less energy, so a weak return is more likely noise regardless of its geometry. IDSOR normalises the distance statistic by range and modulates the threshold by intensity. Its shape parameters were grid-searched on the *validation* split only.

Temporal+SOR Keeps only points with support in the previous sweep, exploiting the fact that real structure persists between 100 ms sweeps while backscatter does not [3]. Ego-motion is compensated by ICP first — see Section 7, correction 3.

Three are learned, trained on paired noisy/clean patches:

PointNet-AE An autoencoder (0.83M parameters) with a global bottleneck and a fully-connected set decoder, trained with Chamfer loss.

DGCNN A lightweight graph network (0.22M parameters) with two heads: per-point displacement regression and per-point outlier classification. The k-NN graph is built once from input coordinates and reused across EdgeConv layers, which halves latency relative to the dynamic feature-space variant.

RangeRestoreNet Developed here; see Section 3.4.

3.4 RangeRestoreNet

The DGCNN's accuracy comes at 1.8 s per frame, three orders of magnitude outside the real-time budget, and the bottleneck is graph construction rather than arithmetic. Following the representational choice the recent literature converged on, the scan is projected into a 64×2048 range image with five channels (range, intensity, x, y, z):

$$u = W\left(1 - \frac{\arctan_2(y, x) + \pi}{2\pi}\right), \qquad v = H\left(1 - \frac{\arcsin(z/r) - f_{\text{down}}}{f_{\text{up}} - f_{\text{down}}}\right).$$

A lightweight U-Net (1.94M parameters) then produces two per-pixel outputs: a noise logit (removal, as the literature does) and a bounded range residual (restoration, which it does not). Downsampling is applied to the azimuth axis only in the first stage, since a 64-row image is already shallow and halving height early destroys the vertical laser structure that separates a surface from a floating droplet.

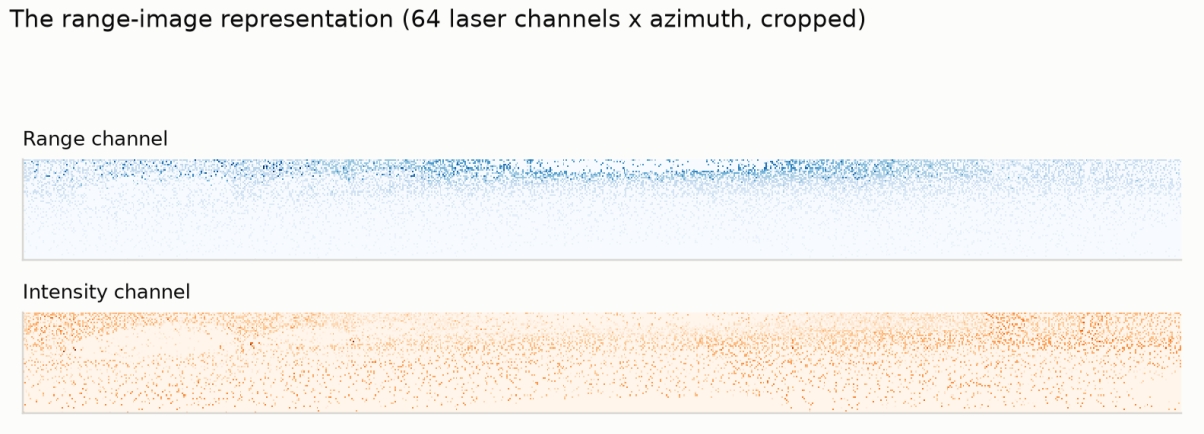


Figure 1: The range-image representation, cropped to 800 azimuth columns. The range channel shows the scene's geometric structure as a depth map; the intensity channel is what distinguishes a weak droplet return from a solid surface at the same distance. Recovering the sensor's native 2-D topology is what allows ordinary convolutions to replace per-patch k-NN graph construction.

Projection is lossy: at 64×2048, 83.3% of points win a pixel, and 92.8% recover their own label after a projection/unprojection round trip (the figure exceeds the representation rate because a point losing a collision inherits its occluder's label, which is usually correct). That 92.8% is the ceiling on this method's removal accuracy and is reported rather than assumed. Voxel downsampling is the cause — it replaces the sensor's 64 discrete laser rings with grid centroids —

but the shared preprocessed cloud is retained so that all six methods are compared on identical input.

3.5 Evaluation protocol

Four choices make the comparison defensible.

Paired and multi-seed. Every method sees the *identical* corrupted cloud for a given (frame, seed). Results are therefore paired, and every ranking claim is supported by a Wilcoxon signed-rank test. Geometry uses 10 frames × 3 seeds (30 paired observations); detection uses **28 frames × 3 seeds (84 paired observations)**, a distinction whose necessity is explained in Section 7.

An oracle floor, not zero. The *oracle* deletes every injected point and keeps every real one — a perfect noise classifier. It scores CD 0.104 m, *not* zero, because the weather destroyed 31% of returns. Each method's *oracle gap*,

$$\text{gap} = 100 \times \frac{\text{CD}_{\text{method}} - \text{CD}_{\text{oracle}}}{\text{CD}_{\text{noisy}} - \text{CD}_{\text{oracle}}},$$

is the fraction of the *recoverable* error it leaves behind. 0% means it matched a perfect classifier; 100% means it achieved nothing.

Removal F1 alongside geometry. Chamfer Distance cannot distinguish "removed the noise" from "moved everything slightly". Every method producing an index set into the corrupted cloud also reports noise-removal precision, recall and F1.

HD_{95} over strict Hausdorff. Strict Hausdorff is the maximum over all residual points, so a single survivor near the 60 m range gate pins it near 30 m for every method. It ranked nothing and is retained only in the raw output.

Detection consistency. A KITTI-pretrained PointPillars detector runs on the clean cloud and its boxes become the reference; we report what fraction survive on each condition (recall) and how many spurious boxes appear (precision), matching by BEV centre distance within 1 m. This bounds recoverable perception performance without fabricating ground truth. It is not absolute mAP, and is not presented as such.

4 Results

4.1 Reconstruction quality

Method	CD (m)	HD_{95}	Oracle gap	Removal F1	Retention
Noisy input	0.572 ± 0.111	—	100%	—	—
Oracle (perfect removal)	*0.104 ± 0.003*	*0.353*	*0%*	*1.000*	*68.9%*
DGCNN	**0.111 ± 0.008**	**0.342**	**1.6%**	0.788	78.7%
IDSOR	0.174 ± 0.013	0.827	15.0%	0.665	81.3%
RangeRestoreNet	0.215 ± 0.035	0.459	23.8%	**0.814**	65.6%
SOR+ROR	0.251 ± 0.020	1.529	31.5%	0.628	78.7%
Temporal+SOR	0.317 ± 0.035	2.173	45.5%	0.567	81.4%
PointNet-AE	0.521 ± 0.092	3.353	89.1%	0.000	100.0%

Table 1: Reconstruction quality over 30 paired observations at 25 mm/h. Every pairwise CD difference is significant at $p < 10^{-5}$ (Wilcoxon signed-rank).

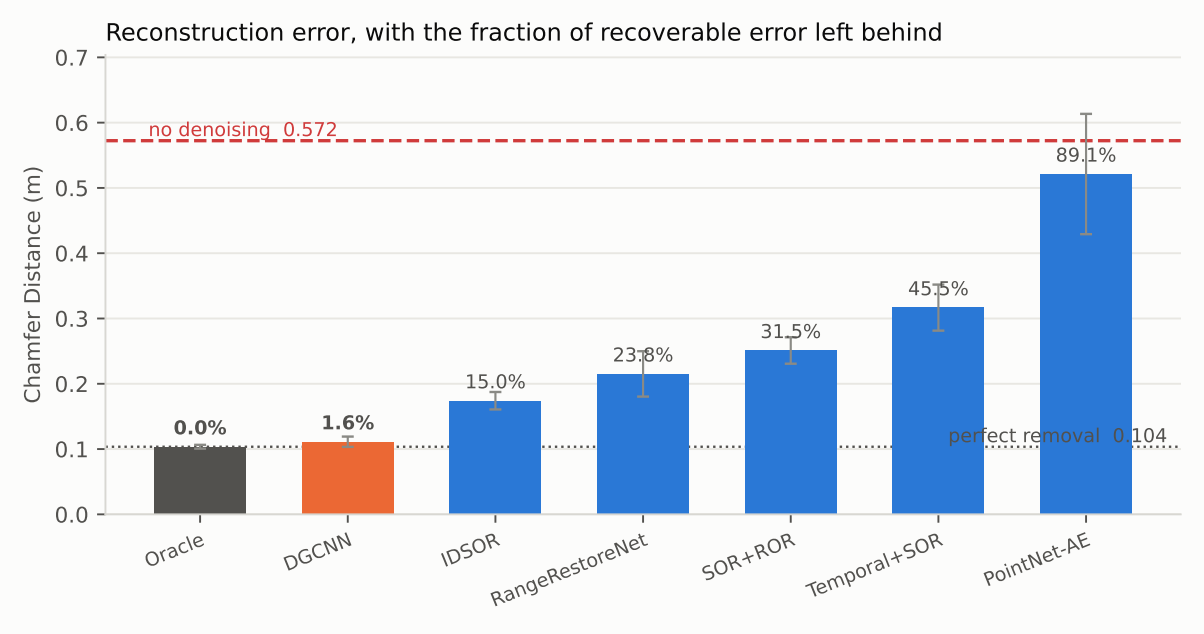


Figure 2: Chamfer Distance by method. Percentages above each bar give the oracle gap. The dotted line is the perfect-removal floor at 0.104 m — non-zero because 31% of returns were destroyed beyond recovery — and the dashed line the uncorrected input. Reading a raw CD against zero rather than against this floor systematically understates every method.

The DGCNN closes 98.4% of the recoverable error. IDSOR is the strongest method requiring no training, improving on the proposal's SOR+ROR baseline on every axis simultaneously: better CD (0.174 vs 0.251), higher precision (0.888 vs 0.772), higher retention, and lower latency.

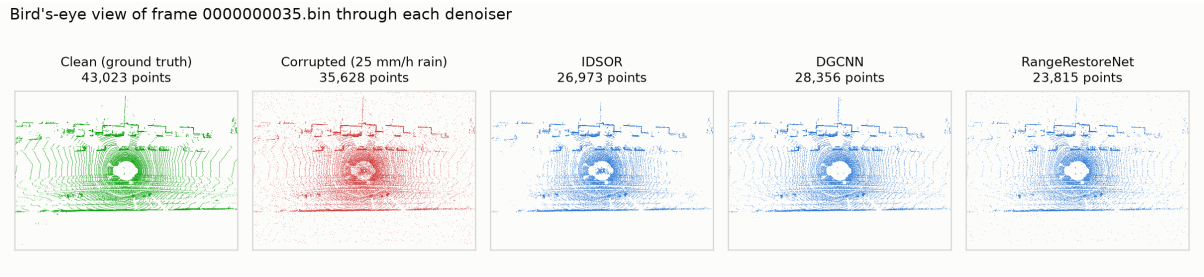


Figure 3: One test frame through each denoiser. The corrupted panel shows the characteristic near-range speckle of backscatter. Note the point counts: the autoencoder is omitted here because it removes nothing at all, so its panel is visually identical to the corrupted input — a fact more informative than any figure of it would be.

4.2 Noise-removal accuracy

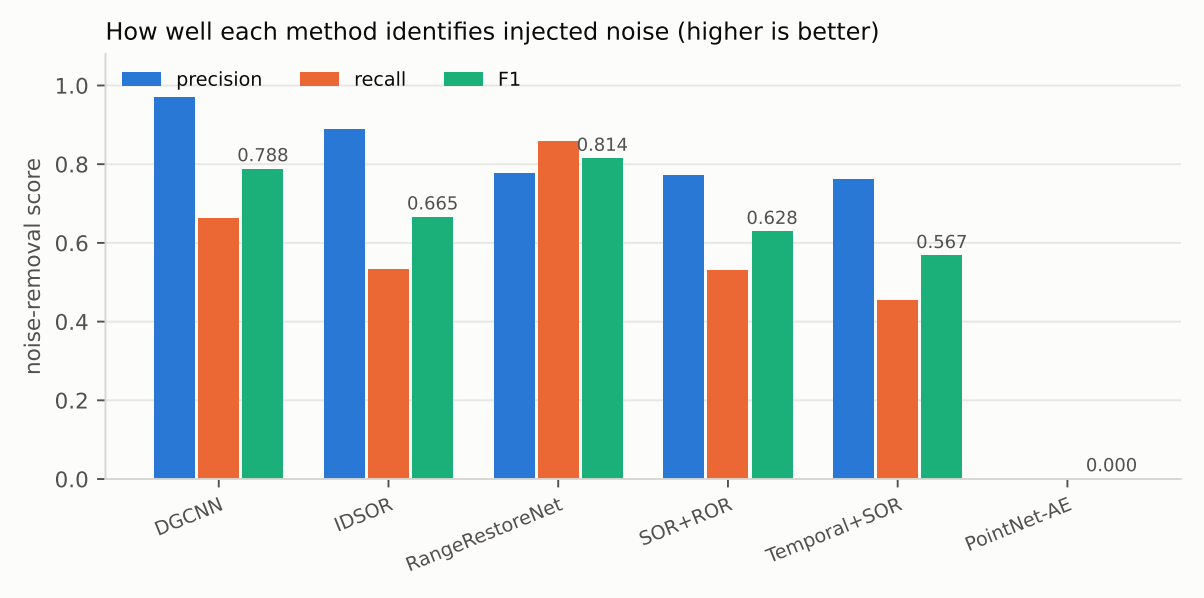


Figure 4: Precision, recall and F1 of noise identification. The two learned classifiers separate along an interpretable axis: the DGCNN is precision-oriented (0.969) and conservative, RangeRestoreNet is recall-oriented (0.859) and aggressive. The autoencoder scores exactly zero because it has no removal head — it is a pure displacement model, which Section 4.3 shows to be actively harmful.

4.3 Downstream detection impact

Condition	Recall	Precision	p vs oracle
Noisy input	0.259 ± 0.231	0.806	$< 10^{-4}$
Oracle (perfect removal)	*0.479 ± 0.325*	*0.805*	—
DGCNN	**0.452 ± 0.298**	0.764	0.153 (n.s.)
RangeRestoreNet	0.403 ± 0.320	0.788	2×10^{-4}
SOR+ROR	0.392 ± 0.303	0.737	$< 10^{-4}$
Temporal+SOR	0.389 ± 0.269	0.795	—
IDSOR	0.368 ± 0.291	**0.826**	$< 10^{-4}$
PointNet-AE	0.062 ± 0.104	0.773	$< 10^{-4}$

Table 2: Detection consistency over 84 paired observations.

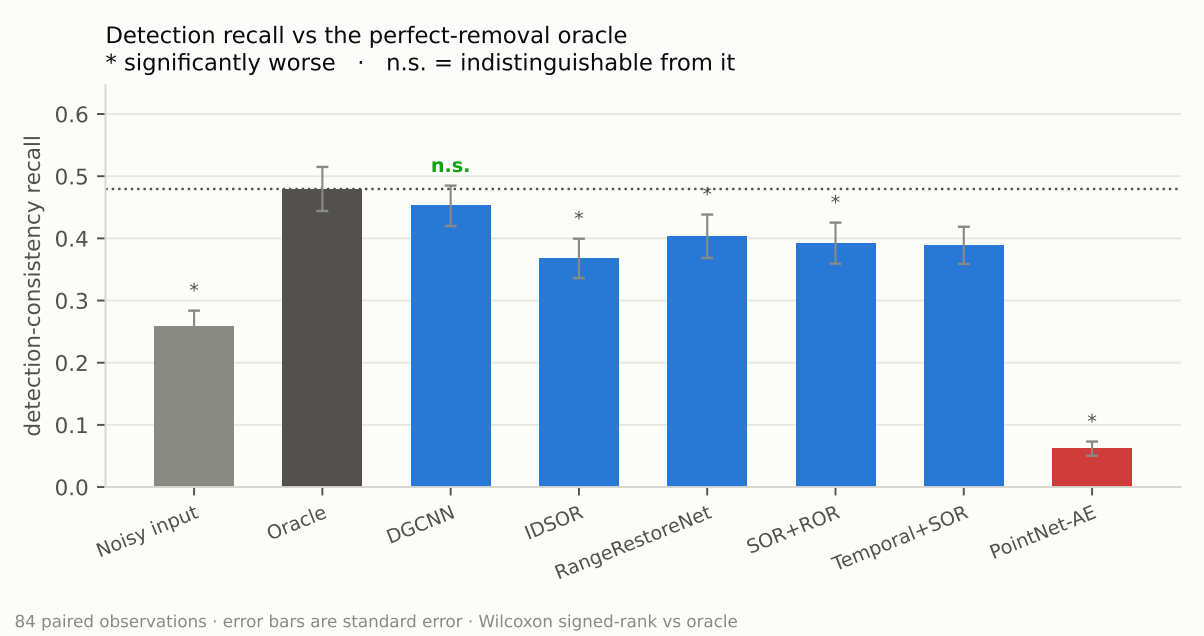


Figure 5: Detection-consistency recall. The DGCNN is the only method statistically indistinguishable from the perfect-removal oracle; all others fall significantly short. The corruption costs the detector 74% of its clean-cloud detections, and every denoiser recovers a significant share except the autoencoder, which makes matters significantly worse.

Perfect removal is the ceiling. Nothing exceeds the oracle. This is worth stating explicitly because an earlier revision of this work concluded the opposite, on the strength of means from ten frames without a significance test (Section 7).

Reconstruction error does not predict perception quality. The autoencoder improves Chamfer Distance by 8.9% relative to the corrupted input while dropping detection recall from 0.259 to 0.062 — a 76% loss at $p < 10^{-4}$, and the only condition worse than no denoising at all. The effect cannot be dismissed as point-count inflation: after the recombination correction of Section 7 the autoencoder emits exactly 100% of the input cardinality. It smooths scan-line

structure into a surface the detector cannot parse, and Chamfer Distance rewards it for doing so.

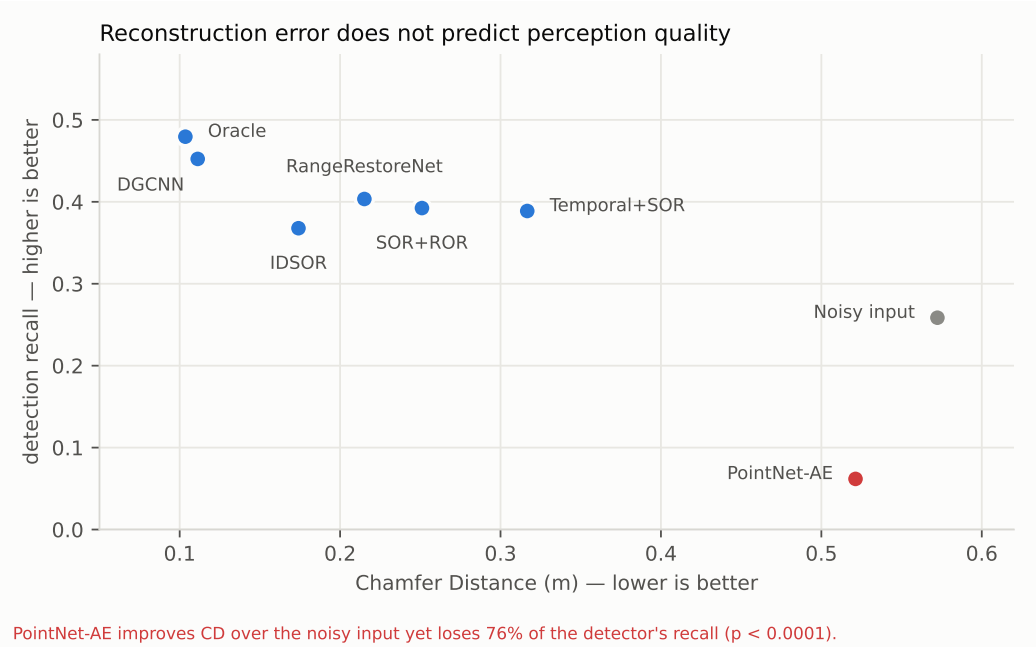


Figure 6: The central result. If Chamfer Distance were a good proxy the methods would lie on a tight descending line. Across all six the rank correlation is only $\rho = -0.60$ ($p = 0.21$) — directionally right, statistically weak — and the autoencoder is a catastrophic outlier: better CD than the corrupted input, near-zero detection recall. A second, milder inversion is visible in IDSOR, which has clearly better CD than SOR+ROR yet slightly lower recall ($p = 0.069$).

Chamfer Distance is improvable by densification. A second and independent demonstration comes from the recombination ablation (Figure 11, left): the uncorrected variant achieves a *better* CD (0.525 vs 0.569) while returning **232% of the input points**. Adding points shortens the clean→denoised nearest-neighbour term faster than it lengthens the reverse term. Chamfer Distance can thus be improved both by blurring and by duplication, neither of which helps perception — which is why every table in this report carries a retention column.

4.4 Real-time feasibility

Method	ms/frame	Hz	≥10 Hz
RangeRestoreNet (MPS)	**45**	**22.1**	**yes**
IDSOR (CPU)	**49**	**20.3**	**yes**
SOR+ROR (CPU)	**65**	**15.5**	**yes**
Temporal+SOR (CPU, incl. ICP)	278	3.6	no
PointNet-AE (MPS)	1,192	0.84	no
DGCNN (MPS)	1,806	0.55	no
DGCNN (CPU, int8)	11,593	0.09	no

Table 3: End-to-end latency from a NumPy array, including format conversion.

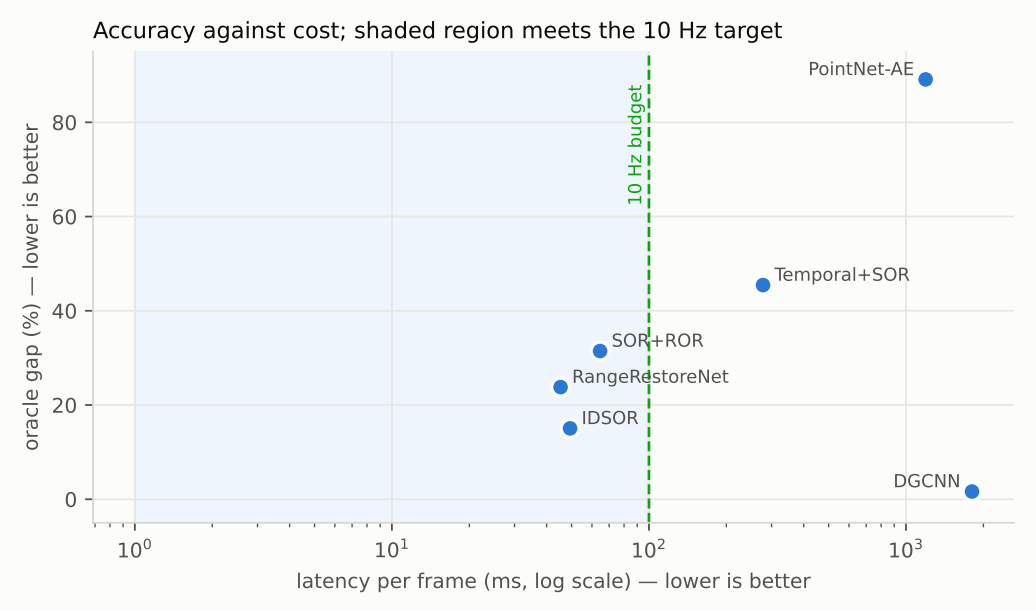


Figure 7: Accuracy against cost. The shaded region meets the 10 Hz target. RangeRestoreNet occupies the position the project needed: within the budget while remaining a learned method.

The 10 Hz target is met, and for the first time by a learned model. RangeRestoreNet runs end-to-end — projection, inference, unprojection — in 45 ms, a 40× speedup over the DGCNN, achieved entirely by changing representation rather than by shrinking the model.

Two caveats keep the classical numbers honest. Part of SOR+ROR's headroom is supplied by the corruption itself: precipitation destroys 31% of returns, so the cloud reaching the filter is smaller than a clear-weather scan, and the same pipeline is slower in good conditions — when it is least needed. And all timings are measured end-to-end including the format conversion that an earlier revision excluded from the baseline's timer.

Dynamic int8 quantisation reduced the autoencoder checkpoint 3.4× (3.34→0.98 MB), which matters for the proposal's embedded Raspberry Pi scenario, but produced no speedup: PyTorch's dynamic backend accelerates Linear layers, whereas the bottleneck is the k-NN gather.

4.5 Training behaviour

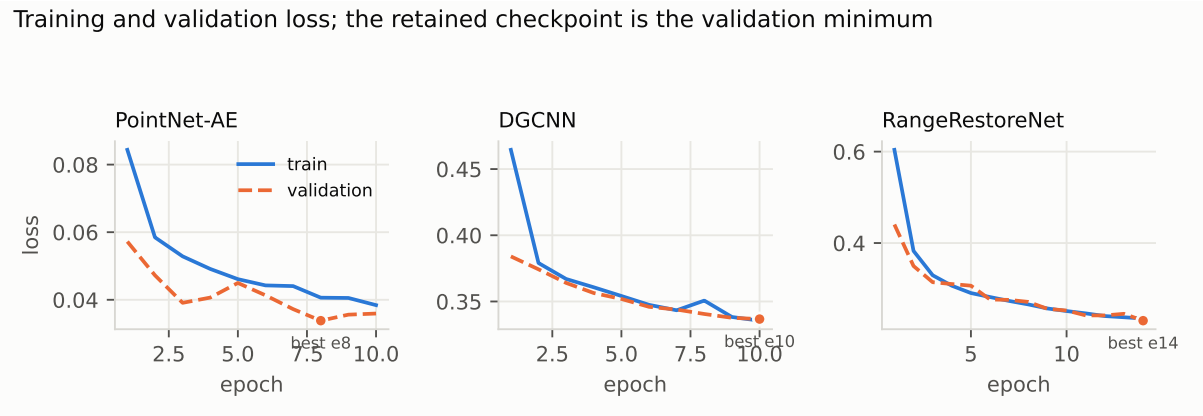


Figure 8: Training and validation loss. The retained checkpoint is the validation minimum, not the final epoch — this is not cosmetic: the autoencoder's validation loss bottoms at epoch 8 and rises thereafter, so last-epoch selection would have shipped a measurably worse model. The DGCNN's validation loss was still descending at epoch 10, so its results should be read as a floor on what the architecture achieves rather than a converged ceiling.

4.6 Ablations

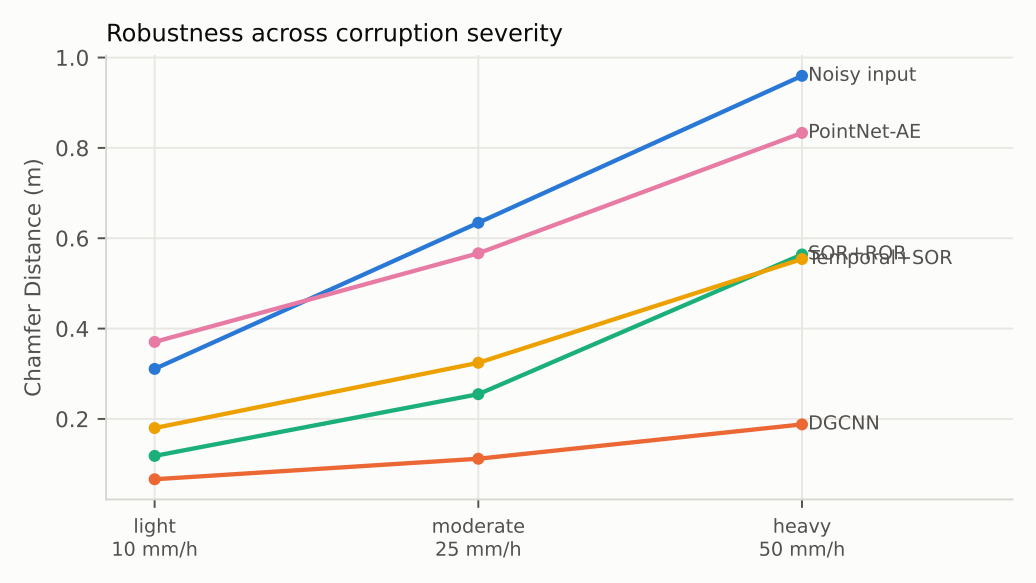


Figure 9: Behaviour across corruption severity. Methods degrade in parallel rather than crossing over, so the ranking is not an artefact of the chosen operating point.

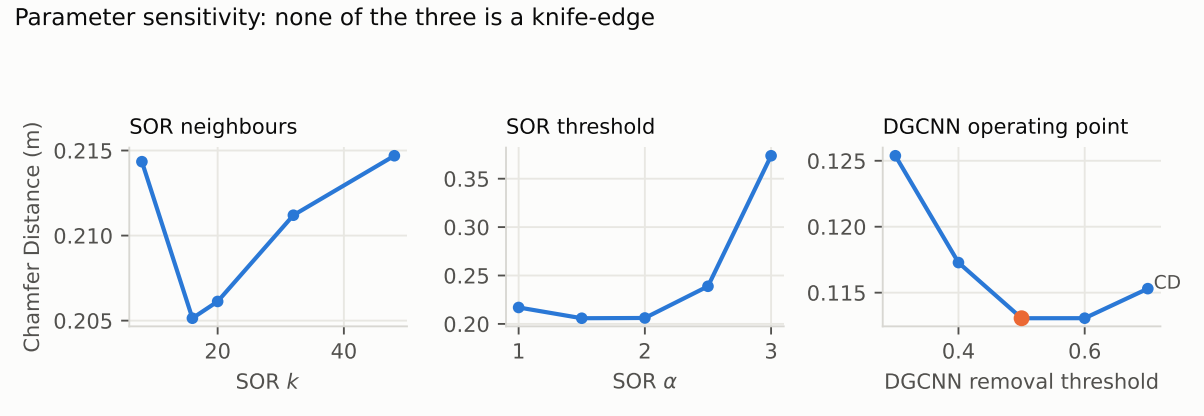


Figure 10: Parameter sensitivity. None of the three parameters is a knife-edge; the DGCNN's removal threshold in particular is flat between 0.5 and 0.6, so the default is not a tuned artefact.

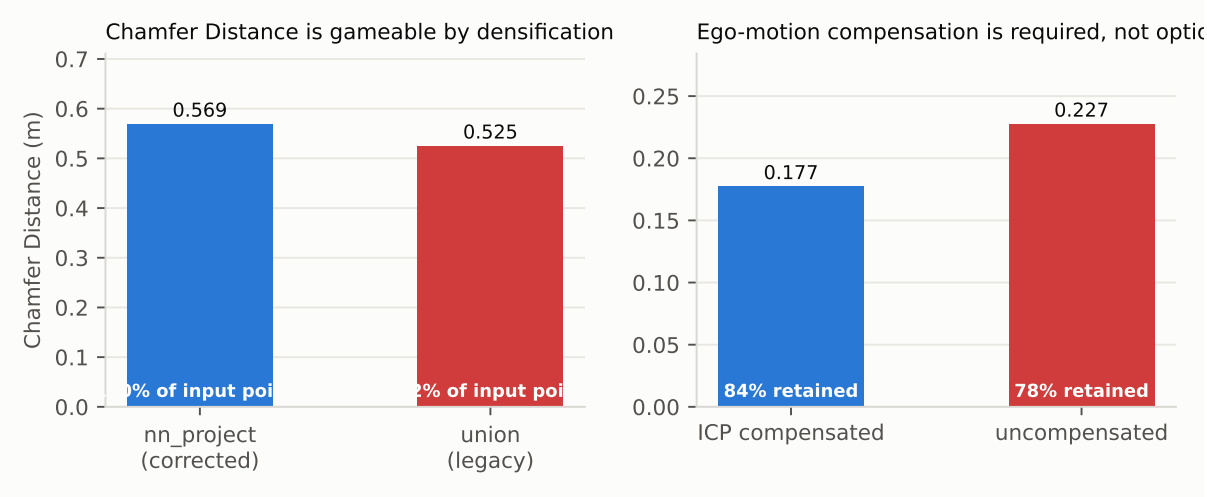


Figure 11: Two ablations that exist to test whether a correction mattered. *Left:* the uncorrected recombination scores a better Chamfer Distance while emitting 232% of the input points — direct evidence that the metric rewards densification. *Right:* ICP ego-motion compensation improves the temporal filter from CD 0.227 to 0.177 while retaining 5 percentage points more of the cloud, confirming the uncompensated variant was deleting static structure rather than backscatter.

5 Supervision Without Ground Truth

The sharpest objection to everything above is that the supervision is circular: we inject the noise, so of course we can identify it. Every learned model here trains on the corruption model's own label array, which no deployed system possesses — and which cannot be obtained from real data, since the same scene cannot be recorded simultaneously with and without rain.

Following LIORNet [5], pseudo-labels are derived from the physics of a weather return using only a single corrupted scan:

1. **Range-conditioned intensity anomaly.** Received power falls as ρ/r^2, so a surface at 40 m legitimately returns less than one at 10 m. A global intensity threshold would merely relabel "far" as "noise", so the median intensity is estimated per range bin from the scan itself and a return is anomalous only relative to its own range.

2. **Neighbourhood range coherence.** A real surface is continuous in the range image; a droplet floats in front of what it occluded, so its neighbours disagree with it.

3. **Edge protection.** Genuine object silhouettes also produce range discontinuities, but coherent ones persisting across adjacent pixels.

LIORNet's fourth rule uses zero-reflectivity, which Ouster sensors report and the KITTI HDL-64E does not, so it is omitted rather than approximated. The restoration head is disabled in this configuration: its target is the clean range of a known source point, which is ground truth by construction.

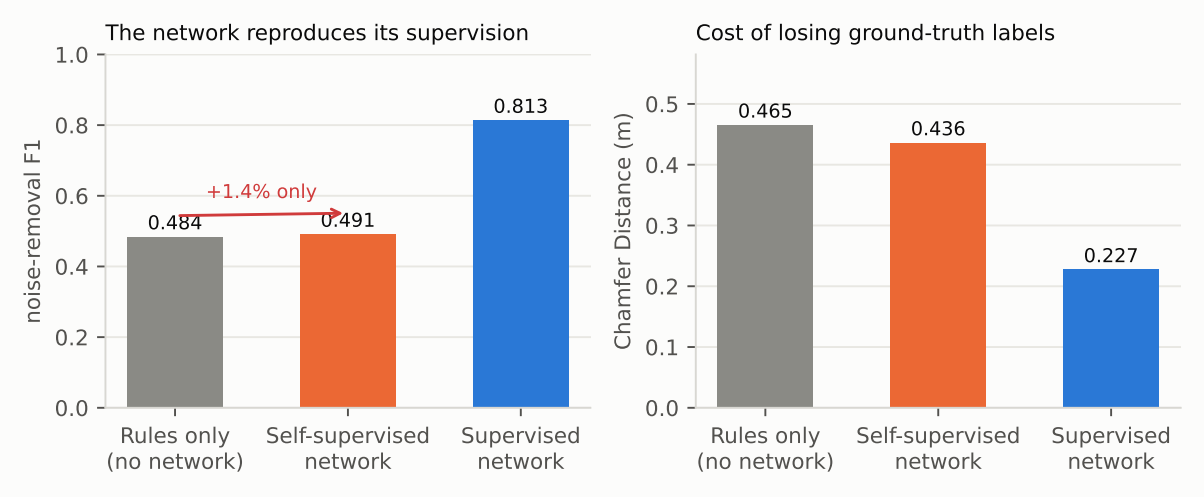


Figure 12: Self-supervision against its own supervision. The pseudo-labels score F1 0.484 against the true labels — roughly half of what the network is told is wrong. Trained on them, the network reaches 60% of the supervised model's removal F1, but moves F1 only from 0.484 to 0.491.

The network reproduces its supervision rather than surpassing it. Reporting "retains 60% of supervised performance" without the rules-only baseline beside it would credit the network for what three physical heuristics already deliver. Learning does help in one respect — HD_{95} improves from 3.297 to 2.564, so the network catches the most damaging surviving outliers better than the rules — but average classification accuracy is unchanged.

This is a negative result, and a useful one: the gap between F1 0.49 and 0.81 measures exactly what knowledge of the ground truth is worth on this task. No additional training on these rules closes it. The prerequisite for progress is better pseudo-labels — multi-echo returns, temporal consistency across successive sweeps, or the reflectivity channel this sensor does not provide — not longer training.

6 Negative Results

Position restoration contributes nothing. Both learned models carry a displacement head, and neither does useful work:

	Mean position error of surviving real points
Corrupted input	0.0804 m
After DGCNN displacement	0.0783 m (−2.6%)
After RangeRestoreNet restoration	0.0804 m (no change)

Removing the DGCNN's displacement entirely leaves Chamfer Distance unchanged (0.1158 vs 0.1157). This is not undertraining. Weather range jitter is **zero-mean**, so given a single corrupted observation the error-minimising prediction *is* zero — the restoration head converges there correctly. Recovering the displacement requires a surface prior over neighbouring returns, which neither a per-point patch MSE nor a per-pixel image L_1 induces.

The practical consequence for the literature question raised in Section 2 is that the survey's observation — that removal-only methods cannot reposition points — is true but does not bind here. Perfect removal remains the ceiling, and a displacement head bolted onto a removal network does not move it. Anyone extending this work should treat position restoration as an open problem requiring an explicit smoothness or surface-consistency term.

7 Experimental Validity

Seven defects affecting reported numbers were identified during review. They are recorded here rather than silently fixed, because several bear directly on the conclusions and one reversed a headline claim.

1. **Autoencoder patch recombination.** Unordered patch reconstructions were concatenated and voxel-downsampled, returning 165% of the input point count — inflating the Chamfer term and handing the detector a cloud of the wrong density. Each input point now adopts its nearest reconstructed point within its own patch.
2. **Autoencoder training target.** The Chamfer target was the nearest-clean-point projection of *every* noisy point, outliers included, so roughly 15% of the target set was surface points duplicated wherever outliers fell. The target is now a genuine clean patch.
3. **Temporal filter ego-motion.** Inter-sweep translation was assumed negligible against the 0.5 m consistency radius. Measured by ICP it averages 0.56 m and reaches 0.94 m, exceeding the radius on half of all frames; uncompensated, the filter deletes static structure rather than backscatter.
4. **Patch coverage.** Randomly seeded patch centres left 2,147 of 36,033 points (6%) outside every patch. Such points retained an outlier probability of zero and could never be removed, capping the DGCNN's removal recall independently of classifier quality. Coverage is now completed greedily, at a cost of 27% more patches (+38 ms/frame) — a latency regression accepted deliberately.
5. **Filter timing.** SOR and ROR started their timers after the NumPy-to-Open3D conversion while learned methods were timed end-to-end, flattering the baseline.

6. **Degenerate frames.** An aggressive filter can empty a sparse frame; the metrics returned infinity, which propagated silently into averaged tables. Such frames now yield NaN and are counted explicitly.
7. **Detection claims made from means with no significance test.** See below.

The most instructive mistake. Detection was originally evaluated on 10 frames. Two methods showed a higher mean recall than the perfect-removal oracle, and an earlier revision of this work concluded that removal alone was not the ceiling and attributed the effect to position correction. Neither part survived scrutiny. Per-frame detection recall has a standard deviation of ≈ 0.3 about a mean of 0.4, so at $n = 10$ a gap of 0.05 between means is uninformative; re-run at 28 frames $\times$ 3 seeds the ordering *reverses*, and both methods are significantly *worse* than the oracle. The proposed mechanism was independently disproved by direct measurement (Section 6).

The geometric metrics never had this problem — their between-frame variance is an order of magnitude smaller — which is precisely why the same sample size was wrongly assumed adequate for both. The general lesson is that statistical power must be established per metric, not per project.

8 Limitations

Stated plainly, since each bounds what the results can be taken to show.

- **The noise is synthetic.** LISA is a physically motivated approximation, not measured adverse weather. Two things mitigate this — the model derives from published Mie-scattering measurements rather than being tuned by us, and the detection evaluation uses a detector that never saw our noise model — but validation on measured data (STF, CADC, WADS) has not been performed and is the single most valuable next step.
- **Detection is consistency, not accuracy.** It measures agreement with the detector's own clean-cloud output, bounding recoverable performance without certifying absolute accuracy. The raw split has no ground-truth boxes.
- **The dataset is small and homogeneous** — 262 scans from two drives of a single urban daytime session. Generalisation across sensors, cities and lighting is untested.
- **Models are not converged.** The DGCNN's validation loss was still descending at epoch 10 on laptop hardware.
- **The intensity channel is idealised.** Our model assigns droplet returns low intensity by construction; real sensors exhibit more complex intensity behaviour, which would weaken IDSOR and the pseudo-labels.
- **The hardware component is deferred.** The proposal's YDLIDAR X2 and Raspberry Pi live-capture element requires the physical sensor. The pipeline is sensor-agnostic and accepts any $(N, 3)$ or $(N, 4)$ scan.

9 Conclusion

Six denoising methods were implemented and compared on a statistically controlled protocol, with reconstruction quality, noise-removal accuracy, downstream detection impact and latency all measured.

The strongest reconstruction comes from a graph network that closes 98.4% of the recoverable error, but at 0.55 Hz. The strongest practical result is *RangeRestoreNet*, which meets the 10 Hz target as the first learned method here to do so, with the best noise-removal accuracy of any method, by adopting the range-image representation the recent literature converged on. Among methods needing no training, IDSOR improves on the proposal's baseline on every axis at once.

The most consequential finding is methodological rather than architectural: **reconstruction metrics do not predict downstream perception quality**. A model can improve Chamfer Distance while destroying three quarters of a detector's recall, and the metric can be improved by duplicating points. Any denoising work evaluated on reconstruction error alone should be read with that in mind.

Two negative results are reported at equal length. Position-restoration heads do not work on zero-mean range jitter, for reasons intrinsic to the estimation problem. Self-supervision from physics-derived pseudo-labels reproduces its supervision rather than improving on it, and the resulting gap quantifies what ground-truth labels are worth. Neither outcome is what was hoped for at the outset, and both are more useful recorded than omitted.

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